

Research Project Report

Twitter Sentiment Analysis using Supervised Machine Learning on TweetEval Dataset

1. INTRODUCTION

With the rapid growth of social media platforms like Twitter (now X), sentiment analysis has become an essential tool for understanding public opinion. Tweets are short, informal, and often noisy, making sentiment classification a challenging task. Traditional lexicon-based approaches struggle with such data, while supervised machine learning models have demonstrated better adaptability and performance [1], [2].

This project focuses on performing **binary sentiment analysis (positive and negative)** using the TweetEval dataset. Multiple machine learning models are implemented and compared to evaluate their effectiveness in handling real-world tweet data.

2. DATASET DESCRIPTION

The **TweetEval dataset** was used in this study, accessed via the HuggingFace library. It originally contains three sentiment classes: positive, neutral, and negative.

For this experiment:

- Neutral tweets were removed to perform **binary classification**
- Labels were mapped as:
 - Positive → 1
 - Negative → 0

After preprocessing:

- Total samples: **24,933 tweets**
- Positive: **17,841**
- Negative: **7,092**

Since the dataset was imbalanced, the **SMOTE (Synthetic Minority Oversampling Technique)** was applied to balance the classes before training.

3. TEXT PREPROCESSING

To improve model performance, the following preprocessing steps were applied:

1. Lowercasing of text
2. Removal of URLs, mentions, emojis, punctuation, and non-ASCII characters
3. Normalization of repeated characters
4. Tokenization
5. Stopword removal
6. Lemmatization

These steps help reduce noise and ensure consistent representation of textual data [3], [4], [5].

4. FEATURE ENGINEERING

The cleaned tweets were converted into numerical form using **TF-IDF (Term Frequency-Inverse Document Frequency)**.

TF-IDF:

- Assigns higher weight to important words
- Reduces impact of commonly occurring words
- Produces sparse, high-dimensional feature vectors

This representation works effectively with classical machine learning models [4], [6] .

5. MACHINE LEARNING MODELS

Eight supervised machine learning models were implemented:

- Logistic Regression
- Naive Bayes
- Linear Support Vector Machine (SVM)
- k-Nearest Neighbors (k-NN)
- Decision Tree
- Random Forest
- XGBoost
- Hard Voting Ensemble

The dataset was split into:

- **80% training**
- **20% testing**

Hyperparameter tuning was applied to Logistic Regression and XGBoost to improve performance.

6. EVALUATION METRICS

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

These metrics provide a comprehensive understanding of classification performance.

7. RESULTS AND ANALYSIS

The results show that:

- **Linear SVM achieved the highest accuracy (~84.62%)**
- **Hard Voting Ensemble (~84.40%)** performed similarly
- Logistic Regression showed strong ROC-AUC performance (~0.90)
- Naive Bayes performed moderately well
- k-NN and Decision Tree showed poor performance due to high-dimensional sparse data

These findings indicate that:

- **Linear models perform best for text classification tasks**
- Ensemble methods improve robustness but do not significantly outperform top models
- Simpler models struggle with sparse feature spaces

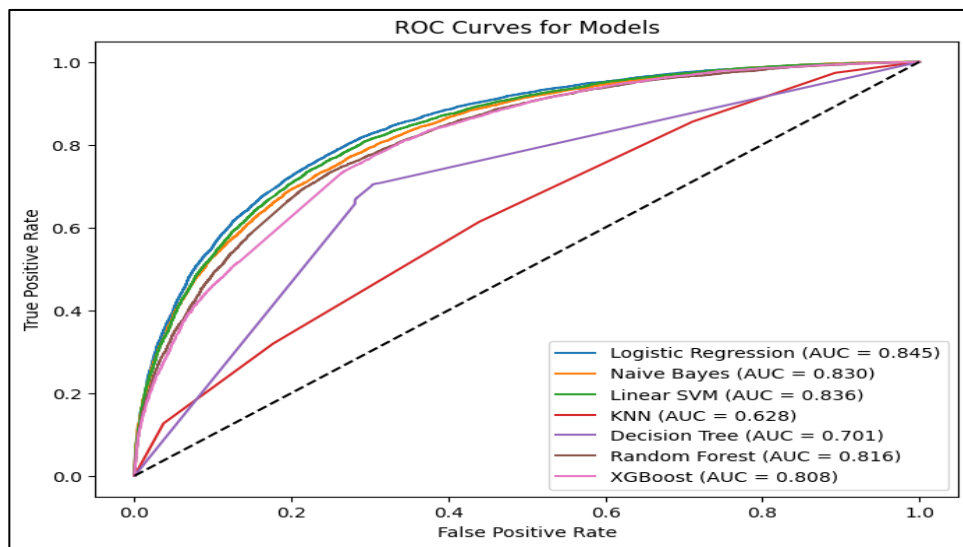


Fig: ROC-AUC performance of ML models on the Sentiment140 dataset

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8406	0.8031	0.8105	0.8066	0.9047
Naive Bayes	0.8207	0.7795	0.8000	0.7880	0.8891
Linear SVM	0.8462	0.8170	0.7947	0.8045	0.9000
k-NN	0.4161	0.6312	0.5822	0.4053	0.6790
Decision Tree	0.7666	0.7122	0.7028	0.7070	0.7028
Random Forest	0.7959	0.7536	0.7200	0.7327	0.8445
XGBoost	0.7957	0.7799	0.6811	0.7026	0.8476
Hard Voting Ensemble	0.8440	0.8091	0.8057	0.8074	—

8. CONCLUSION

This study demonstrates that the TweetEval dataset, being cleaner and more structured, enables better model performance compared to noisier datasets. Among all models, **Linear SVM proved to be the most effective**, achieving the highest accuracy and balanced performance.

Overall, the combination of:

- Proper preprocessing
- TF-IDF feature representation
- Linear classifiers

provides a strong approach for sentiment analysis of Twitter data.

9. REFERENCES / BIBLIOGRAPHY : MENDELEY

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